College of Computer and Information Sciences

CS207

Assignment 1

Computer Sciences Department

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Assignment #1

FOR INSTRUCTOR'S USE ONLY

Q.No.	Max. Marks	Marks Obtained
1	7	
Total Score	7	

General Instructions:

1. Each student has to solve the assignment **individually.**
2. The assignment, after writing the solutions, should be saved as a PDF file and the Design# as a compassed file.
3. The solution should be submitted via the blackboard.
4. Submissions are not allowed after the deadline.
5. Copying solutions from other individuals will result in a mark of ZERO for the assignment.
6. **Assigning a mark to the assignment will only happen after conferring with the student individually.**
7. Due Date: end of Week 6.

Assignment 1

Design a Verilog module along with a block diagram to implement a 4-bit binary adder. The module should take two 4-bit binary inputs, perform the addition operation, and produce a 4-bit result along with a carry-out signal to indicate overflow if it occurs.

Specifications:

1. Inputs:

- A (4-bit binary number)
- B (4-bit binary number)

2. Outputs:

- Result (4-bit binary number representing the sum of A and B)
- Carry_Out (1-bit signal indicating whether a carry-out occurred)

Task:

- Write the Verilog code to implement the adder.
- Draw a schematic diagram of the circuit.
- Simulate the module to verify its functionality using the following inputs:
- A = 1011 (11 in decimal)
- B = 0111 (7 in decimal)

```
 1011
+ 0111
------
10010
```

Expected Output:

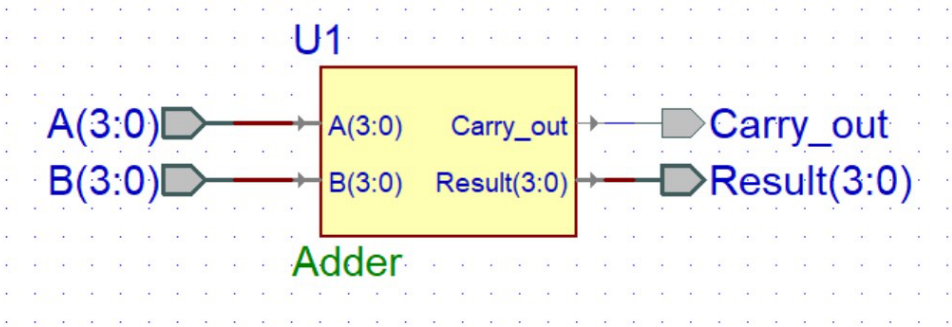
- Determine the values of Result and Carry_Out.

Submission Requirements:

1. Include a screenshot of your Verilog code.
2. Provide a block diagram representing the module.
3. Attach simulation results, including the input signals and output waveform.

Solution

Diagram (Screenshot) (2 Marks):



Code (Screenshot) (2 Marks):

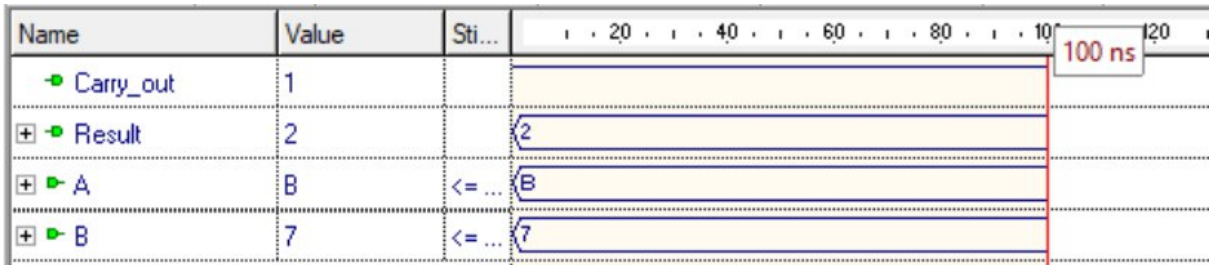
```
//{module {ADDER}}
module Adder(Carry_out, Result, A, B);

output Carry_out;
output [3:0] Result;
input [3:0] A;
input [3:0] B;

assign {Carry_out, Result} = A + B;

endmodule
```

Input and output (Screenshot) (1 Marks):



Brief explanation (Use your own words to write the explanation) (2 Marks):

In this assignment, a 4-bit binary adder was designed using Verilog. The module takes two 4-bit inputs (A and B) and performs a binary addition operation. A block diagram was created to show the relationship between the inputs and outputs. After writing the Verilog code, the design was simulated to verify its functionality. When the inputs A = 1011 and B = 0111 were applied, the output was Result = 0010 and Carry_Out = 1, which indicates that a carry occurred during the addition.